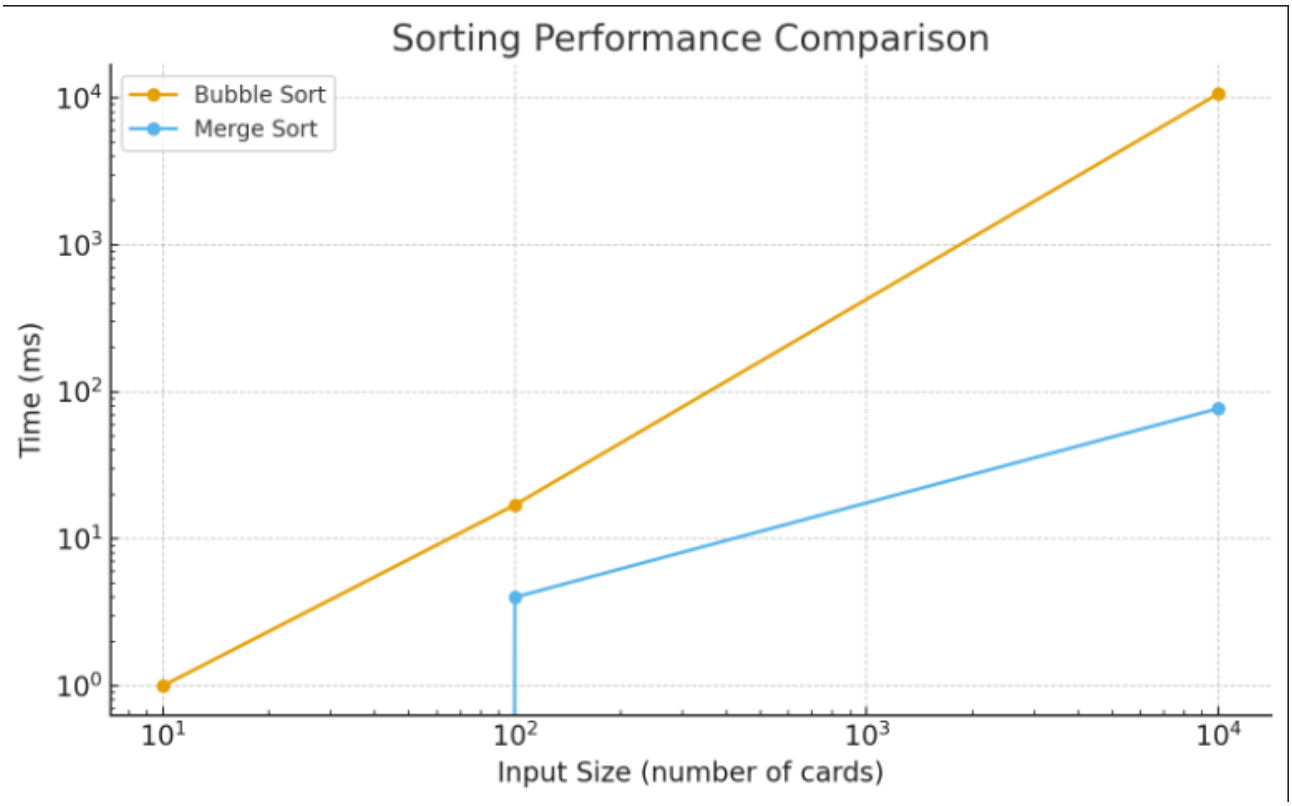
BubbleSort and MergeSort Complexity Reflection



Above is a chart showing the sorting algorithms corresponding with the results of the time taken to sort. Merge sort completed the sorting of 10,100, and 10,000 cards in 0,4 and 77 milliseconds respectively. Bubble sort was able to sort through the 10,100 and 10,000 cards in 1,17 and 10666 milliseconds respectively.

Bubble sorts time to complete increased rapidly responsive to number of cards. This caused it to be much slower than the merge sort towards the end, which is a large drawback to this sort. However, it should be noted that the performance for smaller data sets was nearly identical to the merge sort in terms of performance. Given that bubble sort is much easier to code you could argue for smaller data sets it is more suited compared to merge sort as the small amount of extra time you save from using merge sort isn't worth the time it takes to code the merge sort algorithm.

Merge sort was much better performing with large data sets like the 10,000 values, and it shows as it was 10,589 milliseconds faster, which is very impressive in terms of computer performance. This makes it obviously more suited to larger sets of data. However, as briefly mentioned before, it is much more complex to code with multiple loops and variables needed to complete.

Overall, you could make a case for both algorithms to be better depending on the scenario. As shown by the results bubble sort is much more suited to the smaller data sets and less experienced programmers while merge sort is better for larger data sets and should be used by programmers who have a little more experience.